

House Price Prediction – Task 2

Objective

To improve the House Price Prediction system by applying feature scaling and comparing multiple regression models.

Dataset

California Housing Dataset from scikit-learn.

Steps Performed

1. Loaded the California Housing Dataset.
2. Separated features and target variable.
3. Applied StandardScaler for feature scaling.
4. Split the dataset into training and testing sets.
5. Trained three machine learning models:
6. Linear Regression
7. Ridge Regression
8. Decision Tree Regressor
9. Evaluated model performance using RMSE and R^2 Score.
10. Compared model performance and selected the best model.
11. Visualized Actual vs Predicted House Prices.

Model Comparison

Model	RMSE	R^2 Score
Linear Regression	0.745581	0.575788
Ridge Regression	0.745554	0.575819
Decision Tree Regressor	0.724234	0.599732

Best Model

Decision Tree Regressor achieved the lowest RMSE and highest R^2 Score, making it the best-performing model among the tested models.

Conclusion

Feature scaling and model comparison were successfully implemented. The Decision Tree Regressor provided the best predictive performance for the California Housing Dataset.